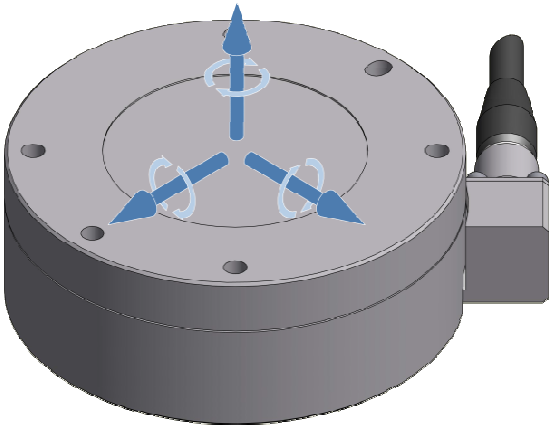
6-Axis Force Torque Sensor

EtherCAT Interface Manual

Revision 0.3

RFT-ECAT-V1



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1. Safety Precaution

1.1. Notices

⚠ Users must thoroughly read and understand all instructions provided in this manual before operating the RFT series F/T sensors. This document details the installation, operation, and technical specifications for RFT series sensors equipped with the EtherCAT interface.

1.2. Warning

⚠ **Inspection:** Do not install or operate any sensor that appears damaged or is missing components.

⚠ **Prohibition of Disassembly:** Unauthorized disassembly or repair is strictly prohibited. Such actions may cause irreparable damage and will void the product warranty.

⚠ **Load Management:** Always calculate the total payload applied to the sensor to ensure safe operation. Exerting force or torque beyond the rated capacity may result in measurement inaccuracies or permanent hardware failure.

⚠ **Environmental Stability:** Ensure the sensor is protected from sudden fluctuations in temperature and humidity. Rapid environmental changes may impair the internal temperature compensation feature, leading to erratic output.

2. Installation and Specifications

2.1. Overview

This manual applies exclusively to RFT models with an **embedded EtherCAT board**. For systems utilizing the external RFTEC-02 board, please refer to the specific "EtherCAT Interface Manual of RFTEC-02".

Item	RFT80-6A02-E	RFT64-6A01-E	RFT80-6A03-E
Product Image			
Temp. Compensation	●	●	●

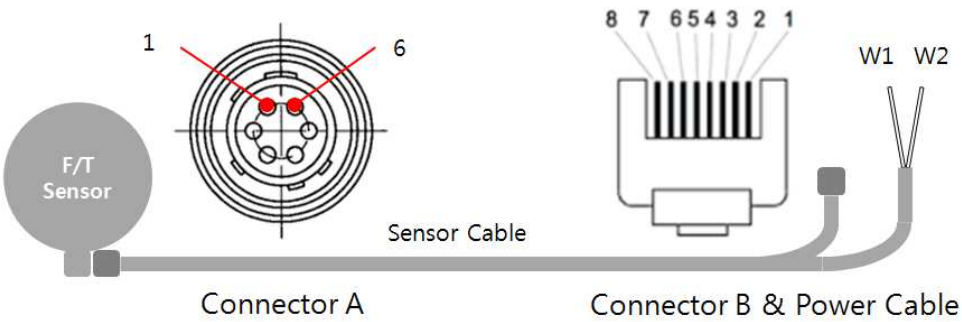
2.2. Electrical Specifications

Item	
Input voltage	5~24Vdc (±10% Tolerance)
Power consumption	Max. 1 W

CAUTION: Exceeding the input voltage limits may cause permanent damage to the sensor. Ensure correct polarity before applying power.

2.3. Wiring and Pin Configuration

The RFT EtherCAT models utilize a specialized cable assembly. Please adhere to the following pinouts:



Connector A (Sensor Side)		
Pin No.	Signal	Color
1	VCC	RED
2	GND	Blue
3	TX +	White
4	TX -	Yellow
5	RX +	Orange
6	RX -	Green

Connector B (Host Side)		
Pin No.	Signal	Color
1	TX+	White
2	Tx-	Yellow
3	Rx+	Orange
4	-	-
5	-	-
6	Rx-	Green
7	-	-
8	-	-

Power Cable (Host Side)		
Pin No.	Signal	Color
W1	VCC	"VCC"
W2	GND	"GND"

- Connector A (Sensor Side): 6-pin circular connector (VCC, GND, TX+/-, RX+/-).
- Connector B (Host Side): RJ45 for EtherCAT communication (TX+/-, RX+/-).
- Power Cable (Host Side): Separate leads for VCC and GND.

3. EtherCAT Bus Interface

The RFT Series EtherCAT interface supports the following core functions:

Real-time Data: High-speed reading of 6-axis force/torque data, overload status, and internal temperature.

Device Identification: Access to model names, serial numbers, and firmware versions.

Configuration: Bias control and adjustable Low-Pass Filter (LPF) settings.

3.1. Process Data Objects (PDO)

PDOs are utilized for deterministic, real-time data exchange.

TxPDO (Object 0x6000): Transmits Force/Torque data and sensor diagnostics to the EtherCAT Master.

RxPDO (Object 0x7000): Currently reserved for future implementation.

3.2. Service Data Objects (SDO) - Dictionary Objects

3.2.1. Object 0x2000: Sensor Information (Read-Only)

Includes Model Name, Serial Number, and Firmware Version as 16-byte strings.

Sub index	Entry Name	Data Type	Description
1	Model Name	STRING(16)	16bytes read-only data for model name
2	Serial Number	STRING(16)	16bytes read-only data for serial number
3	Firmware Version	STRING(16)	16bytes read-only data for firmware version

3.2.2. **Object 0x2001 : Sensor Setup (Read/Write)**

Sub-index 1 (BIAS Setup): Write non-zero (e.g., 1) to set bias; write 0 to un-bias.

Sub-index 2 (LPF Setup): Sets the cut-off frequency (1~500 Hz). Set to 0 or 1000 for raw (unfiltered) data.

Sub index	Entry Name	Data Type	Description
1	BIAS	UINT16	16-bit read-write access for configuring the sensor's bias settings: **Biasing**: write non-zero value (e.g., 1) **Un-biasing**: To clear the bias, write 0 This setting is not saved in non-volatile memory. The bias will automatically reset to 0 (un-biased) upon power cycling the sensor.
2	LPF Setup	UINT16	16-bit Read-Write data used to configure the cut-off frequency of the internal low-pass filter: **Range**: 1 to 500 Hz. **Special Values**: Write 0 or 1000 to bypass the filter and retrieve raw, unfiltered data. This value is stored in non-volatile memory and is retained after power-off.

3.2.3. **Object 0x2002 : Overload Counter**

Tracks the number of overload occurrences for each axis (Fx, Fy, Fz, Tx, Ty, Tz) up to a maximum count of 255.

Sub index	Entry Name	Data Type	Description
1	Fx	UINT8	Number of overload occurrence of Fx
2	Fy	UINT8	Number of overload occurrence of Fy
3	Fz	UINT8	Number of overload occurrence of Fz

4	Tx	UINT8	Number of overload occurrence of Tx
5	Ty	UINT8	Number of overload occurrence of Ty
6	Tz	UINT8	Number of overload occurrence of Tz

3.2.4. **Object 0x6000: TxPDO (Transmit Process Data Object)**

This object consists of eight entries dedicated to the real-time transmission of 6-axis force/torque measurements, operational status, and internal temperature readings. To ensure data integrity, all entries are defined as Read-Only.

Sub index	Entry Name	Data Type	Description
1	Fx	REAL32	Fx measurement (32-bit float, unit: N)
2	Fy	REAL32	Fy measurement (32-bit float, unit: N)
3	Fz	REAL32	Fz measurement (32-bit float, unit: N)
4	Tx	REAL32	Tx measurement (32-bit float, unit: Nm)
5	Ty	REAL32	Ty measurement (32-bit float, unit: Nm)
6	Tz	REAL32	Tz measurement (32-bit float, unit: Nm)
7	FTS-Status	UINT32	This 32-bit unsigned integer indicates the real-time overload status for each axis. Overload Status (1=Overloaded | 0=Normal) Bit 5(Fx) | Bit 4(Fy) | Bit 3(Fz) Bit 2(Tx) | Bit 1(Ty) | Bit 0(Tz) Bit6~31: reserved
8	Temperature	REAL32	Internal Temperature measurement (32-bit float, unit: ℃)

3.2.5. **Object 0x7000: RxPDO (Receive Process Data Object)**

This object is currently **reserved for future implementation** and is not accessible to the user.

4. Testing the EtherCAT Adaper Using TwinCAT

4.1. ESI (EtherCAT Slave Information) Configuration

To ensure proper communication between the TwinCAT Master and the RFT sensor, the ESI XML file must be placed in the designated TwinCAT installation directory.

ESI File Name: RFT-ECAT-V1.xml

Target Directory: C:\TwinCAT\3.1\Config\Io\EtherCAT

Operating Environment: Windows 10 with TwinCAT Version 3.1 or higher.

4.2. Operational Test Sequence

Follow the steps below to initialize the sensor and verify data transmission using the TwinCAT environment.

Step 1: **Project Initialization** – Launch Visual Studio and create a new TwinCAT project.

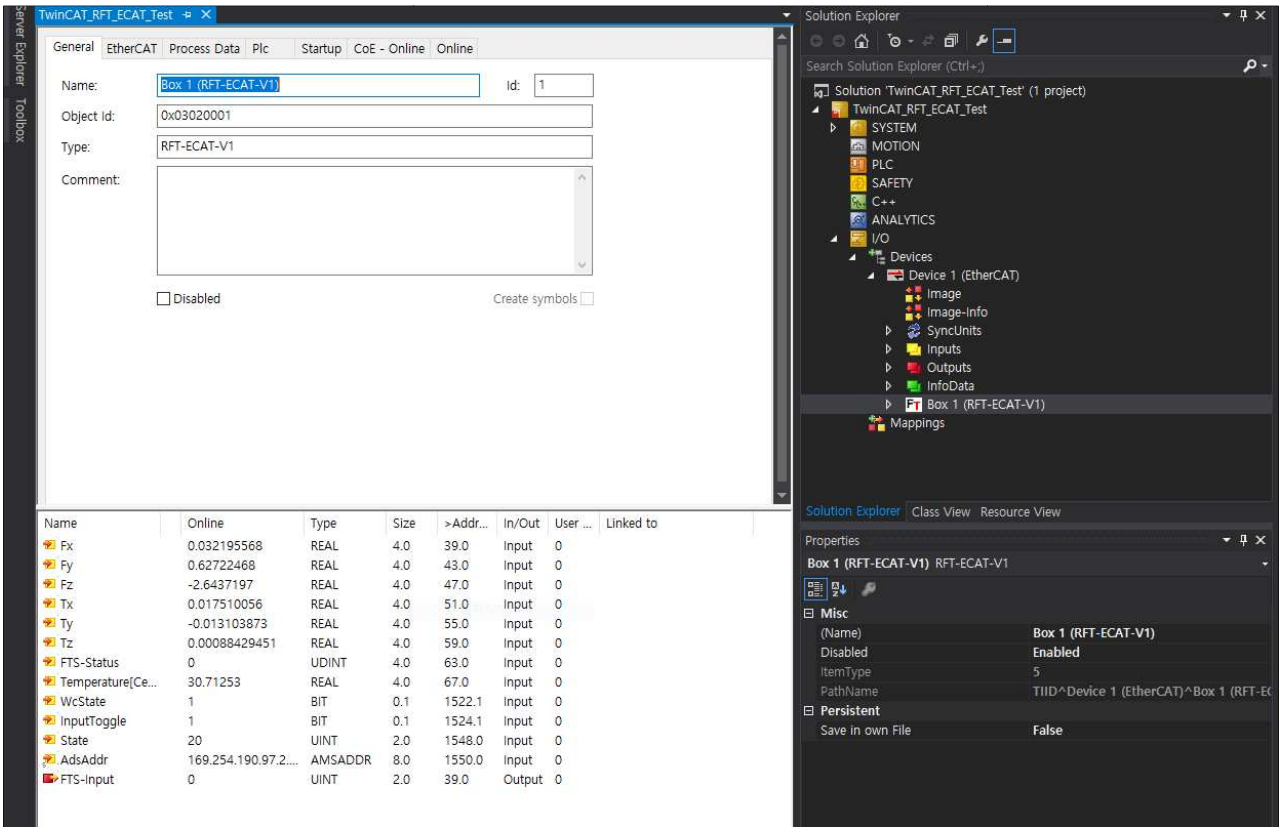
Step 2: **Hardware Connectivity** – Connect a standard LAN cable from the EtherCAT Master (Ethernet adapter) to the sensor.

Step 3: **Power-On** – Apply power to the RFT sensor, ensuring it meets the specified voltage requirements.

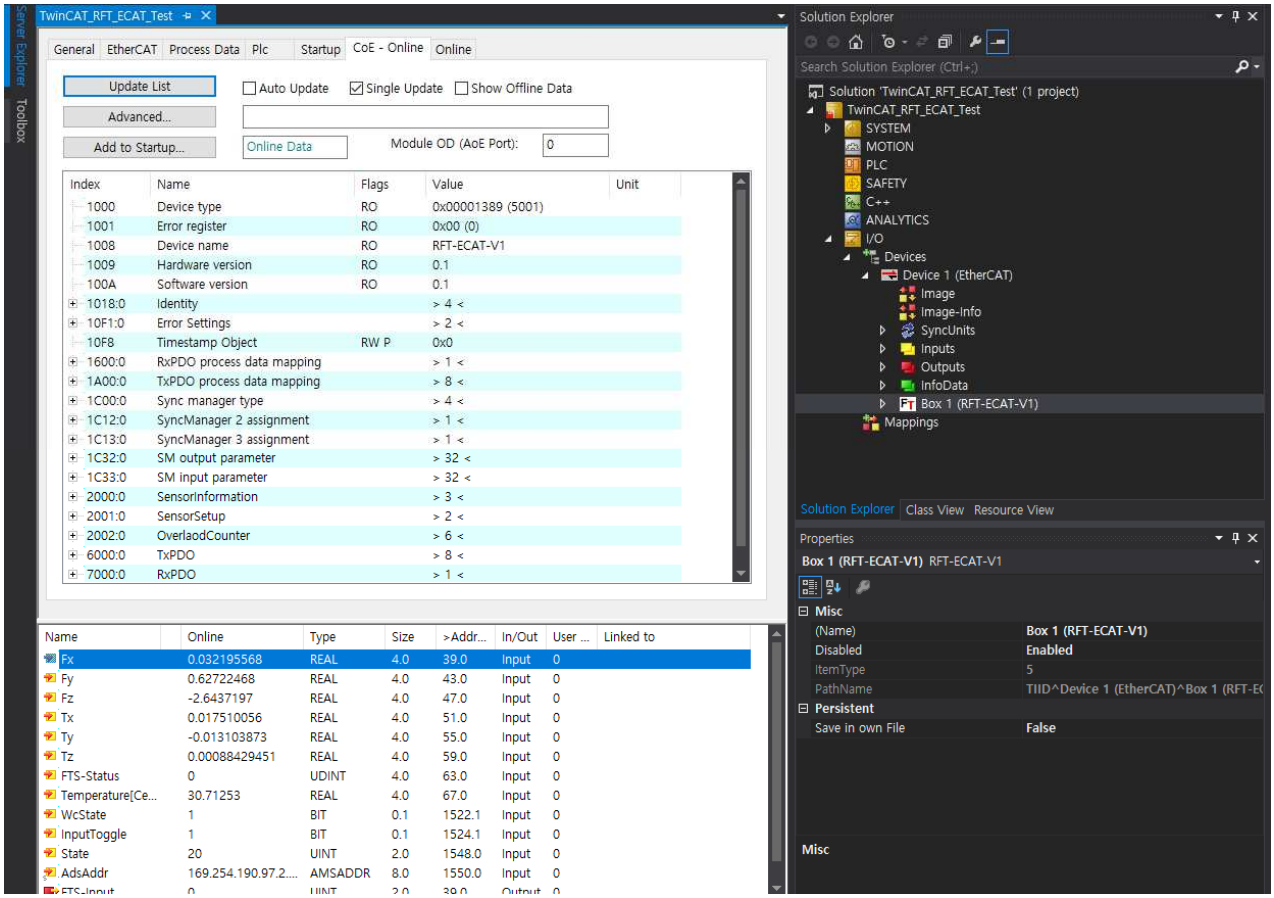
Step 4: **Device Discovery** – Use the "Scan" function within TwinCAT to detect the connected EtherCAT slave devices.

Step 5: **System Activation** – Set the TwinCAT system to Free-Run mode to enable real-time communication.

Step 6: **Data Verification** – Monitor the I/O tree to observe live output values for both PDO (Process Data Objects) and SDO (Service Data Objects). Verify that the force, torque, and status data are updating correctly.



TwinCAT Interface: PDO Runtime Monitoring



TwinCAT Interface: SDO Object Dictionary Acces

5. Product Specification

5.1. Performance Specifications

Model	Dimensions		Weight	Data rate	Load capacity			Resolution		
	ø	H	w/o cable	Max.	Fx, Fy	Fz	Tx, Ty, Tz	Fx, Fy	Fz	Tx, Ty, Tz
	mm	mm	g	Hz	N	N	Nm	mN	mN	mNm
RFT80-6A03	80	33	322	1,000	1,000	1,000	50	200	200	10
RFT80-6A02	80	30	294	1,000	400	400	20	80	80	4
RFT64-6A01	64	30	175	1,000	200	200	10	80	80	4

Resolution is the standard deviation of each six components of force and torque measured for 10 seconds, the measurement data passed through an internal 1st-order low pass filter with cutoff frequency of 100Hz.

Model	Dimensions		Hysteresis			Overload capacity			Crosstalk		
	ø	H	Fx, Fy	Fz	Tx, Ty, Tz	Fx, Fy	Fz	Tx, Ty, Tz	Fx, Fy	Fz	Tx, Ty, Tz
	mm	mm	%FS			%			%FS		
RFT80-6A03	80	33	2	2	2	300	300	300	1.5	1.5	1.5
RFT80-6A02	80	30	1	1	1	300	300	300	2	2	2
RFT64-6A01	64	30	1	1	1	300	300	300	2	2	2

6. Technical Support

For further assistance, please contact our technical support team:

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- Tel. +82-31-606-9918
- E-mail support@robotous.com
- Address Rm 402-1, B-dong, Bundang Techno Park, 723 Pangyo-ro, Bundang-gu, Seongnam-si, Gyeonggi-do 13516, KOREA